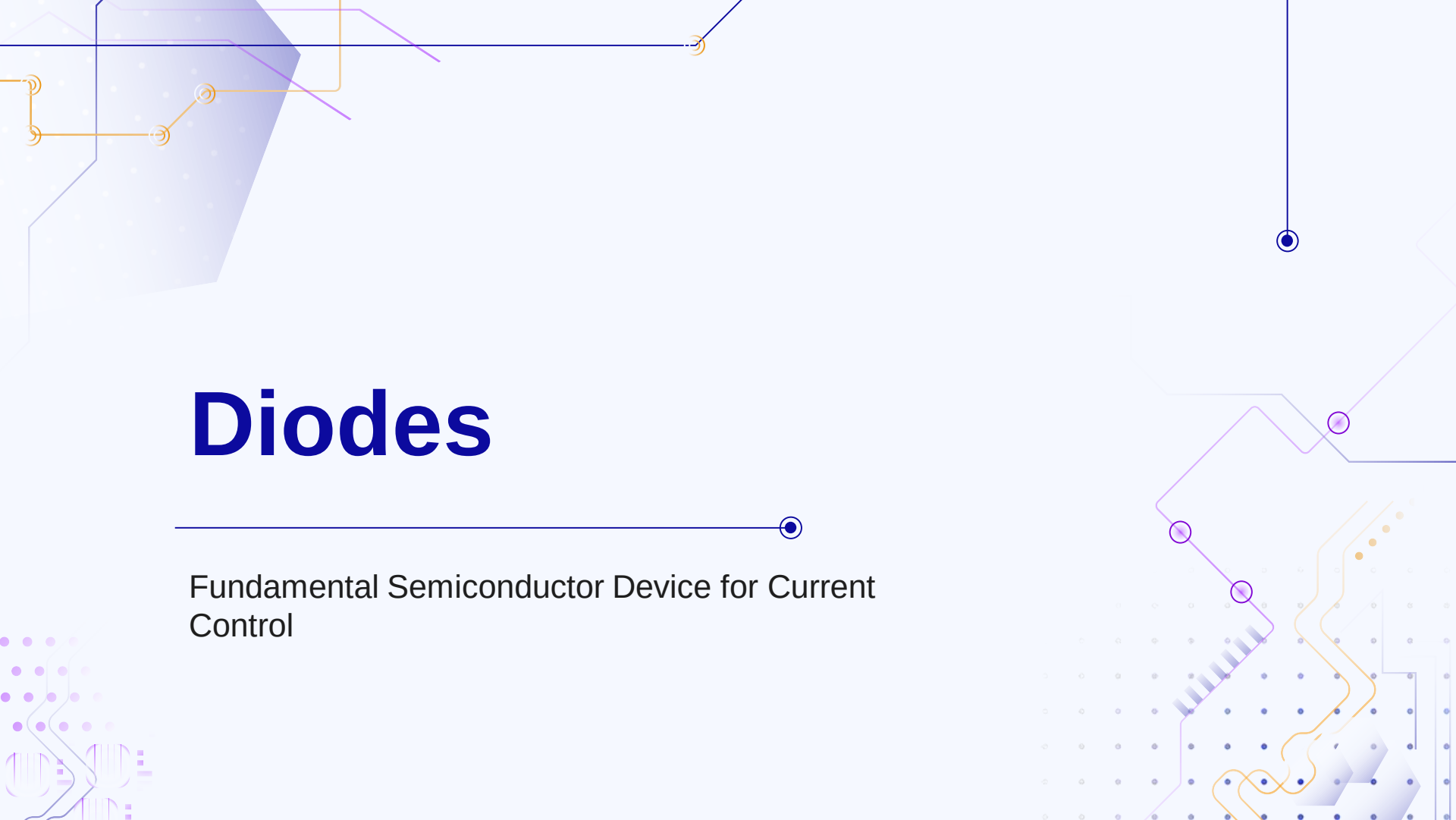




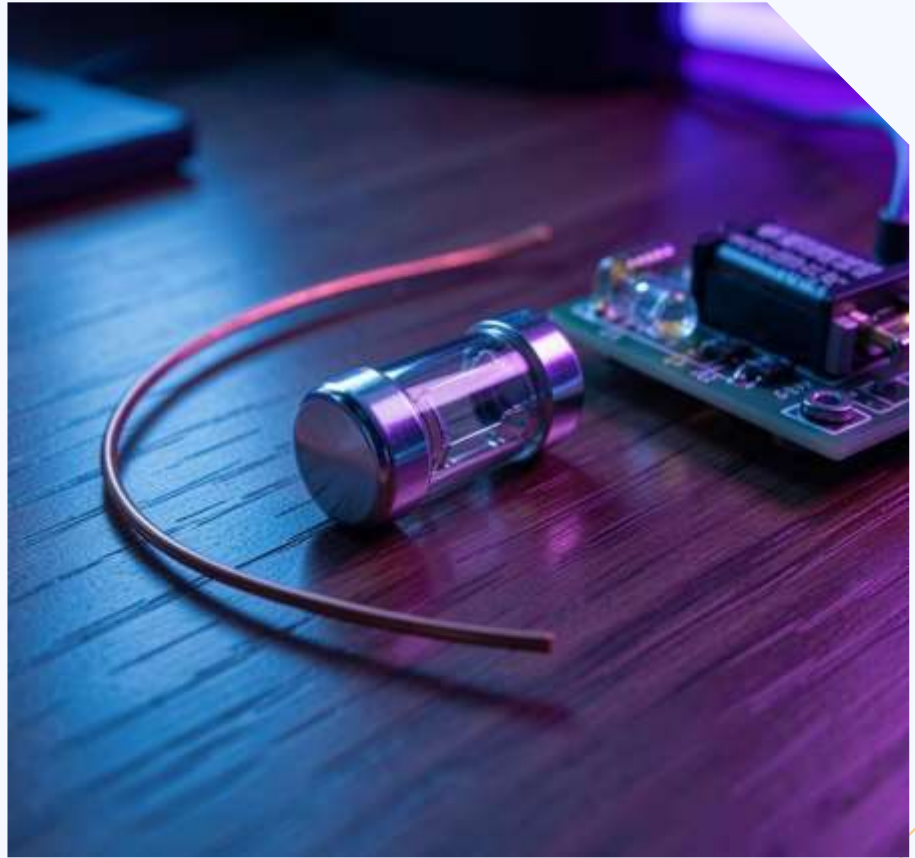
Diodes

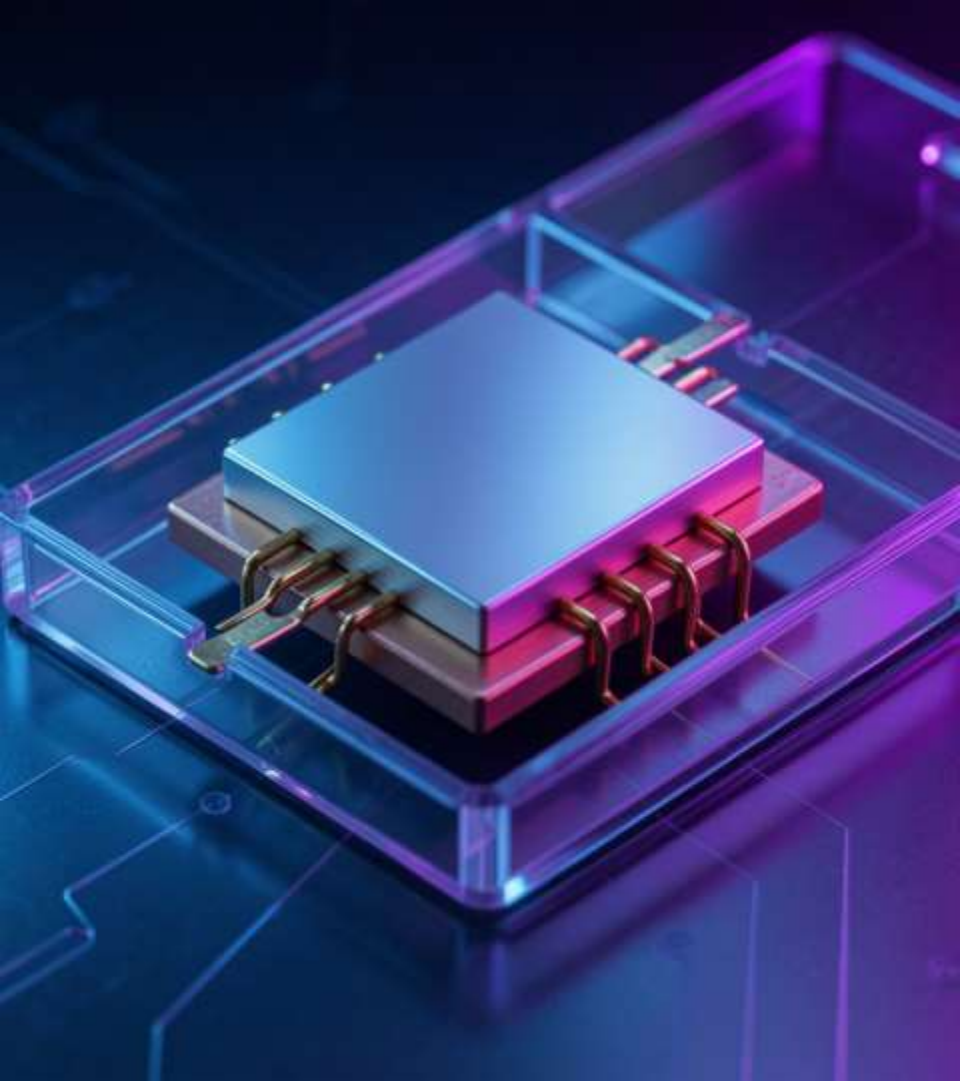
Fundamental Semiconductor Device for Current Control

Introduction

Diodes are two-terminal semiconductor devices allowing current primarily in one direction. They enable *rectification*, *signal control*, and protection in circuits.

Speaker notes: Briefly state diode role in electronics, highlight practical relevance in power supplies and signal processing, and preview topics: structure, V-I behavior, types, and applications.





02

Diode Fundamentals



Definition & Importance

Definition: A diode is a *PN junction* device permitting asymmetric current flow.

Importance: Enables **rectification**, voltage regulation, switching, and protection in electronic systems.

Speaker notes: Emphasize the PN junction concept, practical functions (rectifiers, clamping, LEDs), and why diodes are foundational in circuit design.



History & Development

Early: Vacuum tube and cat's-whisker detectors preceded modern diodes.

Semiconductor era: Development of silicon and germanium PN diodes led to compact, reliable devices and mass adoption in electronics.

Speaker notes: Outline timeline from early detectors to semiconductor diodes, mention key advancements (material shift to silicon) and impact on miniaturization and efficiency.

PN Junction Structure

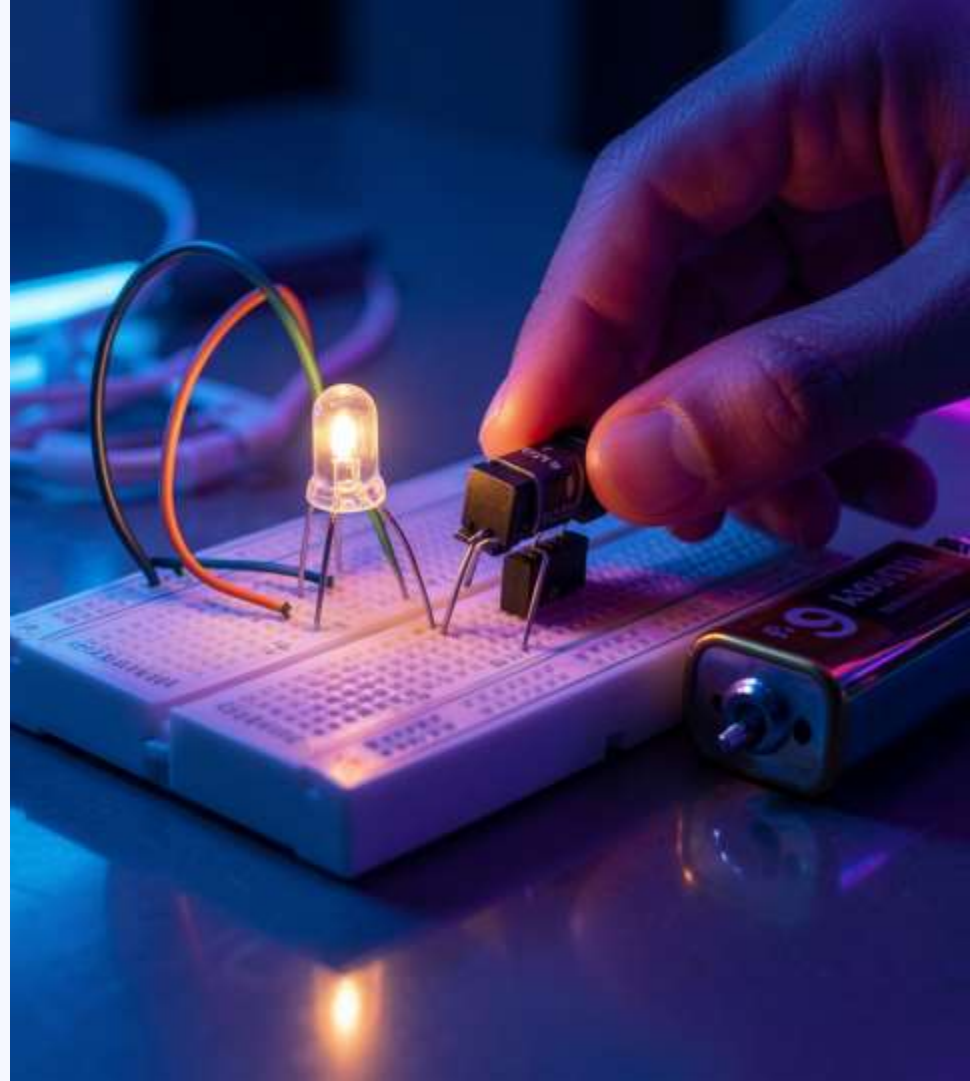
P-type and *N-type* semiconductors form a *PN junction* with a **depletion region** where mobile carriers recombine. Built-in potential prevents charge flow at equilibrium.

Include PN junction diagram and diode symbol.

Speaker notes: Explain doping, carrier diffusion, and formation of the depletion region; point to diagram to show charge distribution and electric field.

02

Diode Behavior & Uses



V-I Characteristics

Diode shows exponential forward I-V; small **forward voltage drop** ($\approx 0.7\text{V}$ Si). In reverse, current is negligible until **breakdown** (Zener/avalanche).

Include V-I characteristic graph.

Speaker notes: Describe forward conduction region, knee voltage, reverse saturation, and breakdown mechanisms; reference graph for operating regions.

Diode Types & Circuits

Rectifier, Zener, LED, Schottky, photodiode — each varies in junction, V_f , speed, and breakdown behavior. Common circuits: half-wave, full-wave, bridge rectifiers.

Include diode symbols for each type.

Speaker notes: Compare properties and typical uses; show schematic snippets of rectifier circuits and note performance trade-offs.

Applications & Modern Importance

Diodes enable **rectification**, voltage regulation, signal clipping, protection, and light emission across power supplies, communications, and renewable systems.

Highlight role in efficient power conversion and protection.

Speaker notes: Emphasize practical impact in modern electronics: SMPS, RF front-ends, solar inverters, and consumer devices; mention trends like wide-bandgap materials.



Conclusions

Diodes are fundamental PN-junction devices controlling current direction, enabling rectification, regulation, switching, and optoelectronic functions. Their varied types meet diverse circuit needs; understanding V–I behavior and junction structure is essential for reliable design.

Speaker notes: Summarize key takeaways, stress practical considerations (V_f , breakdown, speed), and suggest further study: diode modeling and circuit simulation.

Thanks!

- **Do you have any questions?**

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